MATH12345 Scientific Computing
Analysis Exercise 0
The Fibonacci sequence

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The Fibonacci sequence

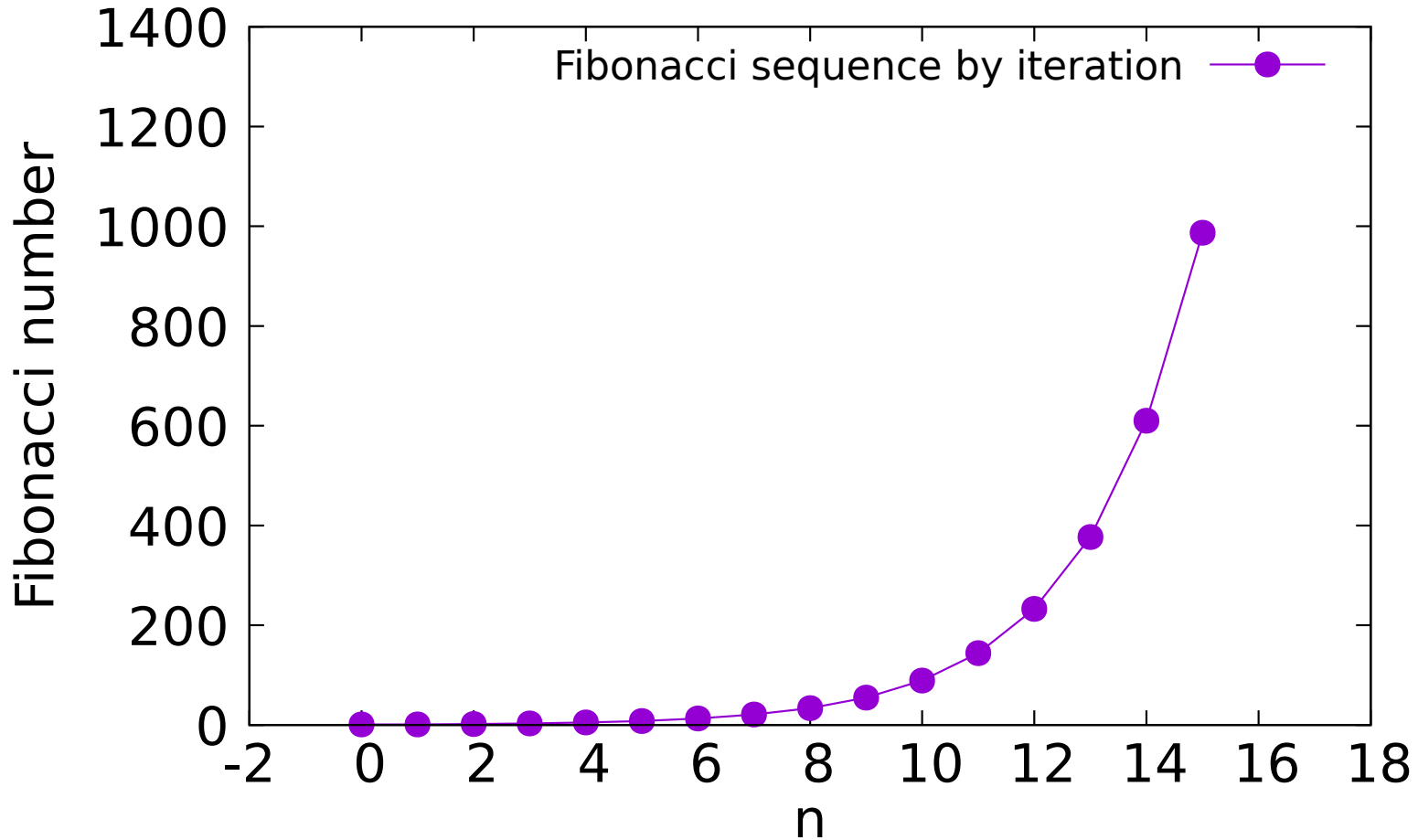
- The Fibonacci sequence is defined by

$$f_0 = 1, \tag{1}$$
$$f_1 = 1, \tag{2}$$
$$f_n = f_{n-1} + f_{n-2} \quad \text{for} \quad n \geq 2. \tag{3}$$

- Aim: compute the sequence f_n $(n = 0, 1, 2, ...)$ by the recursive definition given above and compare against a known explict expression for the n-th value.
- Implementation:
  - Straightforward loop to find f_n, from f_{n-1} and f_{n-2}, following the initialisation of f_0 and f_1.
  - Only need to store f_{n-1} and f_{n-2} in temporary variables. No need to keep track of the entire sequence.

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Result: Plot of f_n vs. n:



The Fibonacci sequence: Validation (i)

A basic validation is to check that each value is the sum of the previous two. Outputting n, f_n for the first few values of n yields

```
0  1
1  1
2  2
3  3
4  5
5  8
6  13
7  21
8  34
9  55
10  89
11  144
```

That seems correct for the first twelve values computed.

[Note: Generally, displaying "raw" numbers or screenshots of a code's output is a bad idea! Use graphical representations.]

The Fibonacci sequence: Validation (ii)

We can also exploit that an explicit expression for the n-th Fibonacci number is known (e.g. using the Wikipedia reference provided):

$$f_n = \frac{\phi^{n+1} - \psi^{n+1}}{\phi - \psi},$$

where $\psi = -1/\phi = (1 - \sqrt{5})/2$. (Note the offset in the index compared to the result given on Wikipedia!)

Result: Plot of f_n vs. n, using the result from the code (filled circle) and from the explicit expression (square): They agree!

